Estimation and Inference for Causal Explainability

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Abstract

Understanding how much each variable contributes to an outcome is a central question across disciplines. A **causal view of explainability** helps uncover underlying mechanisms and generalize to new contexts. Using **semiparametric efficiency** theory, we construct a **one-step estimator** that exploits **treatment independence** and has no larger asymptotic variance than its unrestricted-model counterpart. For **zero total explainability**, we establish an almost-sure partially sharp null and a **randomization-based inference** procedure.

Causal ANOVA

Set-up: We observe n i.i.d. units with K randomized treatments $W_i = (W_{1,i}, \ldots, W_{K,i})$, independent of each unit's potential outcomes. Only $Y_i = Y_i(W_i)$ is observed. Assume $\mathbb{E}[Y^2] < \infty$ and $V := \mathrm{Var}(Y) > 0$. Let $\mu(W) = \mathbb{E}[Y \mid W]$ and $\nu(W) = \mathbb{E}[Y^2 \mid W]$, and write φ for an influence function of the estimand ξ.

Assumptions:

- **(Independent treatments)** The factors W_k, $k \in [K]$, are mutually independent.
- **(Shared additive error)** The potential outcomes satisfy $Y_i(w) = f_Y(w) + E_{Y,i}$, with the same unit-level error across all w, and $E_{Y,i}$ independent of W_i.

Definitions:

- **(Total explainability)** Let W' have the law of W and be independent of $(W, Y(\cdot))$. For $\mathcal{S} \subseteq [K]$,

$$\xi(\vee_{k\in\mathcal{S}} W_k) := \frac{\mathrm{Var}\left(Y(W) - Y(W'_{\mathcal{S}}, W_{-\mathcal{S}})\right)}{2\,\mathrm{Var}\left(Y(W)\right)}.$$

- **(Interaction explainability)** For $k \neq k' \in [K]$,

$$\xi(W_k \wedge W_{k'}) := \frac{\mathrm{Var}\left(I(W'_k, W'_{k'})\right)}{4\,\mathrm{Var}\left(Y(W)\right)},$$

where, for the same unit and with $W_{-\{k,k'\}}$ held fixed,

$$I(W'_k, W'_{k'}) = Y(W'_k, W'_{k'}) - Y(W'_k, W_{k'}) - Y(W_k, W'_{k'}) + Y(W_k, W_{k'}).$$

Identified forms:

- **(Total explainability)** For any $\mathcal{S} \subseteq [K]$,

$$\xi(\vee_{k\in\mathcal{S}} W_k) = \frac{\mathbb{E}[\mathbb{E}^2[Y \mid W]]}{\mathrm{Var}\,(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid W_{-\mathcal{S}}]\right]}{\mathrm{Var}\,(Y)}.$$

- **(Interaction explainability)** For $k \neq k' \in [K]$,

$$\xi(W_k \wedge W_{k'}) = \frac{\mathbb{E}[\mathbb{E}^2[Y \mid W]]}{\mathrm{Var}\,(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid W_{-k}]\right]}{\mathrm{Var}\,(Y)} - \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid W_{-k'}]\right]}{\mathrm{Var}\,(Y)} + \frac{\mathbb{E}\left[\mathbb{E}^2[Y \mid W_{-\{k,k'\}}]\right]}{\mathrm{Var}\,(Y)}.$$

Influence functions

Write $O = (Y, W)$, $\mu = \mathbb{E}[Y \mid W]$, $a = \mathbb{E}Y$, $V = \mathrm{Var}(Y) > 0$, $m_S = \mathbb{E}[Y \mid W_{-S}]$, and $\theta_S = \mathbb{E}[m_S^2]/V$. Then

$$\xi(\vee_{k\in S} W_k) = \theta_\varnothing - \theta_S,$$
$$\xi(W_k \wedge W_{k'}) = \theta_\varnothing - \theta_{\{k\}} - \theta_{\{k'\}} + \theta_{\{k,k'\}}.$$

NIF (orange): unrestricted model. Its efficient influence function is

$$\psi_S(O) = \frac{2Ym_S - m_S^2}{V} - \frac{\theta_S(Y-a)^2}{V}.$$

EIF (light blue): independent treatments, unknown marginals. With unrestricted $P_{Y|W}$ and $P_W = \prod_k P_{W_k}$, the efficient projection is

$$\varphi_S = \psi_S - \mathbb{E}[\psi_S \mid W] + \sum_{k=1}^{K} \mathbb{E}[\psi_S \mid W_k].$$

Use the same linear combinations to obtain ψ^ξ, φ^ξ. Orthogonality gives

$$\mathrm{Var}(\psi^\xi) = \mathrm{Var}(\varphi^\xi) + \mathrm{Var}(\psi^\xi - \varphi^\xi).$$

Both labels denote efficiency in their respective models; equality is possible. The boundedness condition below ensures square-integrable scores.

Cross-fitted one-step estimation

Use i.i.d. observations, fixed K, L, and data-independent folds I_l with $\min_l |I_l|/n$ bounded away from zero. Train outside I_l. For $Q_{-I} = \prod_k \widehat{P}_{W_k,-I}$ and $Q_{S,-I} = \prod_{k\in S} \widehat{P}_{W_k,-I}$, set

$$\widehat{m}_{S,-I}(w_{-S}) = \int \widehat{\mu}_{-I}(w)\, dQ_{S,-I},$$
$$\widehat{a}_{-I} = Q_{-I}\widehat{\mu}_{-I}, \qquad \widehat{V}_{-I} = Q_{-I}\widehat{\nu}_{-I} - \widehat{a}_{-I}^2,$$
$$\widehat{\theta}_{S,-I} = Q_{-I}\widehat{m}^2_{S,-I}/\widehat{V}_{-I}.$$

Here $Qf = \int f\, dQ$ and $\nu = \mathbb{E}[Y^2 \mid W]$. Form $\widehat{\xi}_{-I}$ by the same linear combinations; obtain $\widehat{\varphi}^\xi_{-I}$ using these fitted moments and product law. For $i \in I_{l(i)}$, define

$$Z_i = \widehat{\xi}_{-l(i)} + \widehat{\varphi}^\xi_{-l(i)}(O_i),$$
$$\widehat{\xi} = \frac{1}{n}\sum_i Z_i, \qquad \widehat{\sigma}^2 = \frac{1}{n}\sum_i (Z_i - \widehat{\xi})^2.$$

Sufficient conditions (each fold). Define

$$r_l = \max\left\{\|\widehat{\mu}_{-I} - \mu\|_2, \|\widehat{\nu}_{-I} - \nu\|_2, \max_k \|\widehat{p}_{k,-I} - p_k\|_{2,k}\right\},$$

where $\|\cdot\|_2 = L_2(P_W)$ and $\|\cdot\|_{2,k} = L_2(P_{W_k})$ norms.

(C1) $|Y|, |\mu|, |\widehat{\mu}_{-I}| \le C < \infty$; regression bounds hold on true and fitted supports.

(C2–C3) $r_l = o_P(n^{-1/4})$, hence nuisance consistency.

(C4) Each true/fitted marginal pair has a common dominating measure and the same fixed support, with $p_k, \widehat{p}_{k,-I} \ge c > 0$ a.e.

(C5) A fixed P-Donsker class contains $(p_k - \widehat{p}_{k,-I})^2$ with probability tending to one, for every k, l.

The expansion uses the **true** influence function:

$$\sqrt{n}(\widehat{\xi} - \xi) = \frac{1}{\sqrt{n}}\sum_i \varphi^\xi(O_i; P) + o_P(1)$$
$$\xrightarrow{d} \mathcal{N}(0, \sigma^2), \qquad \sigma^2 = \mathrm{Var}\{\varphi^\xi(O; P)\}.$$

For $\sigma^2 > 0$, a Wald interval is $\widehat{\xi} \pm z_{1-\alpha/2}\widehat{\sigma}/\sqrt{n}$. Define the estimator arbitrarily on zero-denominator events; their probability vanishes.

Randomization Test of Zero Total Explainability

At $H_0 : \xi(\vee_{k\in\mathcal{S}} W_k) = 0$, the total-explainability efficient influence function vanishes, so ordinary first-order Wald intervals do not apply. Under the stated sampling and shared-error assumptions,

$$H_0 \iff Y(W) = Y(W'_{\mathcal{S}}, W_{-\mathcal{S}}) \quad \text{a.s.}$$

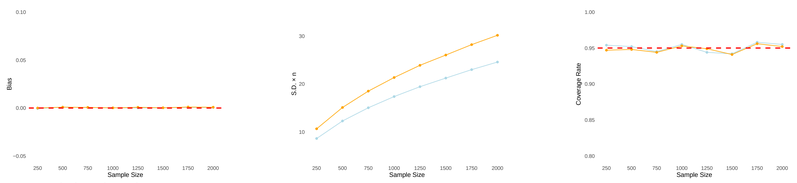
The equality is almost sure over potential outcomes and replacement assignments; for discrete treatments it holds almost surely at each positive-probability assignment pair. Thus supported reassignments satisfy $Y_i(W^*_{\mathcal{S},i}, W_{-\mathcal{S},i}) = Y_i$ almost surely for all i.

For i.i.d. units and independent treatments, draw B permutations independently and uniformly from the full permutation group, including the identity. Permute $W_{\mathcal{S}}$ across units while retaining $(Y, W_{-\mathcal{S}})$. Recompute the same statistic and fitting rule on every dataset, with any auxiliary randomness fixed independently of assignments. Writing T_0 for the observed statistic and T_b for permutation b, use

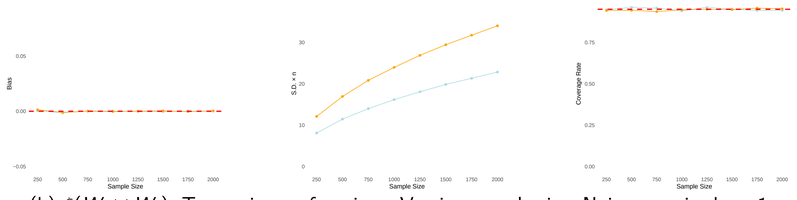
$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{T_b \ge T_0\}}{B+1}.$$

Under H_0, the $B+1$ statistics are jointly exchangeable, so $\mathbb{P}(p \le \alpha) \le \alpha$ for every finite B; weak comparisons handle ties conservatively. For interaction inference, normal intervals require positive variance of the chosen influence function.

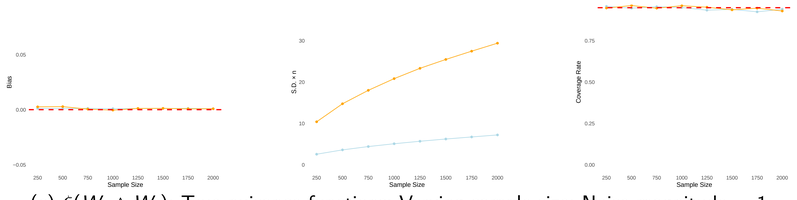
Simulations



(a) $\xi(W_3)$; True nuisance functions; Varying sample size; Noise magnitude = 1



(b) $\xi(W_1 \vee W_3)$; True nuisance functions; Varying sample size; Noise magnitude = 1



(c) $\xi(W_1 \wedge W_3)$; True nuisance functions; Varying sample size; Noise magnitude = 1

Comparison of bias (left panel), estimated standard deviation times sample size (mid panel), and coverage rate (right panel; significance level $\alpha = 0.05$). Independence-model EIF in light blue; unrestricted-model EIF in orange (NIF in the plots). True nuisance functions are used. Results aggregated over 1000 trials.